Data visualization

the grammar of graphics

*the **ggplot2** package, and the five named graphs*

`aes()`

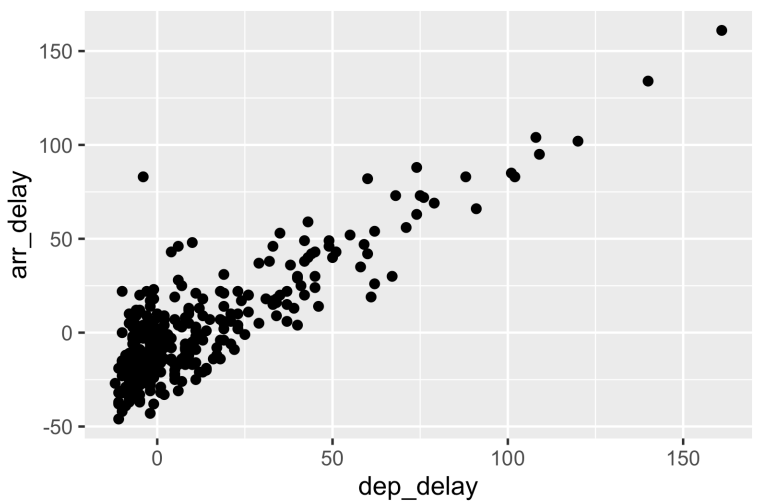
ggplot()
geom_point()
geom_line()
geom_histogram()
geom_boxplot()
geom_bar()

Every graph in this chapter is built from the same three pieces.

ggplot()

Name the data, say which variables to map, then add a **layer** saying what kind of graph to draw.

Data, Aesthetics, Geometry



ggplot()

the verb (function)

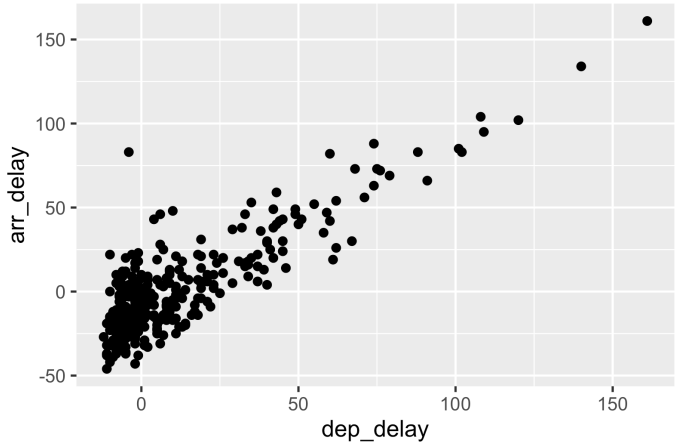
the data frame

which variables to plot

```
ggplot(data = envoy_flights, mapping = aes(x = dep_delay, y = arr_delay)) +
  geom_point()
```

what kind of graph to make

Example



geom_histogram()

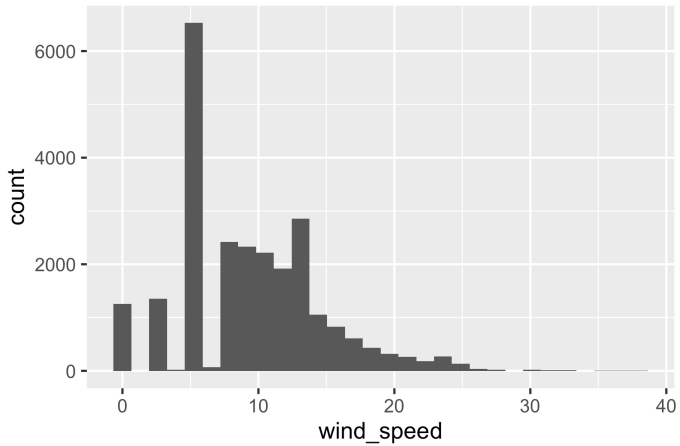
the data frame

which variable to plot

```
ggplot(data = weather, mapping = aes(x = wind_speed)) +
  geom_histogram()
```

what kind of graph to make

Example



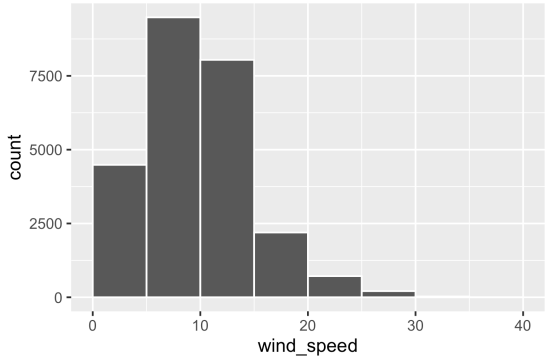
geom_histogram()

Adding **graphical details**. The extra arguments change how the bars are cut and colored, not what is plotted.

```
ggplot(data = weather, mapping = aes(x = wind_speed)) +
  geom_histogram(binwidth = 5, boundary = 40, color = "white")
```

other graphical details

Example



Summary of the five named graphs

	Named graph	Shows	Geometric object	Notes
1	Scatterplot	Relationship between 2 numerical variables	`geom_point()`	
2	Linegraph	Relationship between 2 numerical variables	`geom_line()`	Used when there is a sequential order to the x-variable, e.g. time
3	Histogram	Distribution of 1 numerical variable	`geom_histogram()`	A faceted histogram shows the distribution of 1 numerical variable split by 1 categorical variable
4	Boxplot	Distribution of 1 numerical variable split by 1 categorical variable	`geom_boxplot()`	To graph without splitting by a categorical variable add `x = NULL`
5	Barplot	Distribution of 1 categorical variable	`geom_bar()` when counts are not pre-counted, `geom_col()` when they are	Stacked and dodged barplots show the distribution of 2 categorical variables